Elaina Thomas

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Environmental microbial ecologist integrating metatranscriptomics, field incubations, and culture experiments to understand how marine microbes respond to their environment. My current work focuses on protist-organic matter interactions, coupling field-derived transcriptomes with culture growth and eventual isotope-labeled organic matter experiments to quantify physiological responses and trace organic matter uptake and fate via osmotrophy. I am seeking a postdoctoral position to develop mass spectrometry and metabolomics skills and continue investigating microbe-biogeochemical interactions.

Education

University of Washington Graduate School of Oceanography. Seattle, WA.
PhD in Oceanography, September 2021–present, expected graduation September 2027.
Advisor: E.V. Armbrust.
MS in Oceanography awarded May 2024.
Advanced to PhD candidacy December 2025.
Thesis: Elusive trophic modes of marine protists.

Carleton College. Northfield, MN.
BA in Biology, cum laude, March 2018.
Thesis: Effects of dispersal ability and spatial scale on niche and neutral processes.

Appointments

National Science Foundation Graduate Research Fellow, University of Washington. September 2021–present.
Advisor: E.V. Armbrust
Lead two complementary projects on marine protist trophic modes: (1) analyzed metatranscriptomes from field samples and at-sea incubations and developed machine learning models to predict trophic mode; and (2) currently integrate field metatranscriptomic analyses with culture experiments adding glycine betaine to an axenic phytoplankton, followed by C- and N-labeled glycine betaine additions and targeted metabolomics to trace uptake and metabolic fate.

Senior Research Support Associate, MIT. January 2019–August 2021.
Advisor: S.W. Chisholm
Led and supported bioinformatic analyses for multiple projects, including 16S-based identification of heterotrophs in picocyanobacteria enrichment cultures, analysis of RNA, methylation, and genomic sequencing from dark-adapted *Prochlorococcus*, and characterization of novel integrative elements and their regulation in *Prochlorococcus*.

Research Assistant, Carleton College. January–November 2018.
Advisor: Rika Anderson
Analyzed hydrothermal vent metagenomes to characterize viral community structure, mined CRISPR loci to build virus–host interaction networks, and collected viral metagenomes during the 2018 Lost City Expedition.

REU Student, Eco-Informatics Summer Institute, Oregon State University. June–August 2017.
Advisors: Julia Jones and Rebecca Hutchinson
Examined long-term datasets on plant and pollinator communities in alpine meadows of the Western Cascades to quantify diversity patterns over space and time.

Publications

<u>First author</u>
* =co-first author

Elaina Thomas, Mora J. Groussman, Sacha N. Coesel, Nicholas J. Hawco, Randelle M. Bundy, E. Virginia Armbrust. 2025. Latitude- and depth-driven divergence in protist trophic strategies revealed by a machine learning model. *Front. Microbiol*. 16. 10.3389/fmicb.2025.1602162. Model: https://github.com/armbrustlab/MarPRISM.

Elaina Thomas*, Rika Anderson*, Viola Li, Jenni Rogan, and Julie A. Huber. 2021. Diverse viruses in deep-sea hydrothermal vent fluids have restricted dispersal across ocean basins. *mSystems* 6 (3). 10.1128/mSystems.00068-21.

Sean M. Kearney, **Elaina Thomas***, Allison Coe, and Sallie W. Chisholm. 2021. Microbial diversity of co-occurring heterotrophs in cultures of marine picocyanobacteria. *Environmental Microbiome* 16 (1). 10.1186/s40793-020-00370-x.

<u>Co-author</u>

Allison Coe, Rogier Braakman, Steven J. Biller, Aldo Arellano, Christina Bliem, Nhi N. Vo, Konnor von Emster, **Elaina Thomas**, Michelle DeMers, Claudia Steglich, Jef Huisman, Sallie W. Chisholm. Emergence of metabolic coupling to the heterotroph *Alteromonas* promotes dark survival in *Prochlorococcus*. 2024. *ISME Commun*. 4 (1). 10.1093/ismeco/ycae131.

Giovanna Capovilla, Rogier Braakman, Gregory P. Fournier, Thomas Hackl, Julia Schwartzman, Xinda Lu, Alexis Yelton, Krista Longnecker, Melissa C. Kido Soule, *Elaina Thomas*, Gretchen Swarr, Alessandro Mongera, Jack G. Payette, Kurt G. Castro, Jacob R. Waldbauer, Elizabeth B. Kujawinski, Otto X. Cordero, Sallie W. Chisholm. Chitin utilization by marine picocyanobacteria and the evolution of a planktonic lifestyle. 2023. *PNAS* 120 (20). 10.1073/pnas.2213271120.

Jessie Berta-Thompson, **Elaina Thomas**, Andrés Cubillos-Ruiz, Thomas Hackl, Jamie W. Becker, Allison Coe, Steven J. Biller, Sallie W. Chisholm. Draft genomes of three closely related low-light-adapted *Prochlorococcus*. 2023. *BMC Genomics* 24. 10.1186/s12863-022-01103-4.

Thomas Hackl, Raphaël Laurenceau, Markus J. Ankenbrand, Christina Bliem, Zev Cariani, **Elaina Thomas**, Keven D. Dooley, Aldo A. Arellano, Shane L. Hogle, Paul Berube, Gabriel E. Leventhal, Elaine Luo, John Eppley, Ahmed Zayed, John Beaulaurier, Ramunas Stepanauskas, Matthew B. Sullivan, Edward F. DeLong, Steven J. Biller, Sallie W. Chisholm. 2023. Novel integrative elements and genomic plasticity in ocean ecosystems. *Cell* 186 (1). 10.1016/j.cell.2022.12.006.

William J Brazelton, Julia M McGonigle, Shahrzad Motamedi, H Lizethe Pendleton, Katrina I Twing, Briggs C Miller, William J Lowe, Alessandrina M Hoffman, Cecilia A Prator, Grayson L Chadwick, Rika E Anderson, **Elaina Thomas**, David A Butterfield, Karmina A Aquino, Gretchen L Früh-Green, Matthew

O Schrenk, Susan Q Lang. 2022. Metabolic strategies shared by basement residents of the Lost City hydrothermal field. *Applied and Environmental Microbiology* 88 (17). 10.1128/aem.00929-22

Allison Coe, Steven J. Biller, **Elaina Thomas**, Konstantinos Boulias, Christina Bliem, Aldo Arellano, Keven Dooley, Anna N. Rasmussen, Kristen LeGault, Tyler J. O'Keefe, Eric L. Greer, and Sallie W. Chisholm. 2021. Coping with darkness: the adaptive response of marine picocyanobacteria to repeated light energy deprivation. *Limnology and Oceanography* 66 (9). 10.1002/lno.11880.

Julia A. Jones, Rebecca Hutchinson, Andy Moldenke, Vera Pfeiffer, Edward Helderop, **Elaina Thomas**, Josh Griffin, and Amanda Reinholtz. 2019. Landscape patterns and diversity of meadow plants and flower-visitors in a mountain landscape. *Landscape Ecology* 34 (5). 10.1007/s10980-018-0740-y.

Grants and awards

National Science Foundation Graduate Research Fellowship, $138,000. 2021–2026.
Egtvedt Endowed Scholarship in Oceanography, University of Washington, $7,633. 2023–2025.
Theodore H. and Marie M. Sarchin Endowed Fellowship in Oceanography, University of Washington, $2,487. 2024–2025.
Identity, Belonging and Inquiry in Science (IBIS) Program Research Supplies Award, University of Washington, $500. 2026.
College of the Environment Conference Presentation Travel Award, University of Washington, $1,500. 2026.
Graduate School Conference Presentation Travel Award, University of Washington, $500. 2026.
National Science Foundation-funded Travel Award, Biogeochemical Exchanges at Sea Ice Interfaces (BEPSII) Workshop, Cambridge Bay, Nunavut, $1,662. 2025.
Graduated cum laude, Carleton College. 2018.
2018 Dean's List, Carleton College. Academic year 2016–2017.

Presentations

Elaina Thomas. Trophic response of protists to latitudinal and depth-driven changes in their environment. Virtual and in-person poster. *Simons Collaboration on Ocean Processes and Ecology Annual Meeting*. 2024.

Elaina Thomas. Mixotrophic protists employ diverse strategies in response to nutrients and light across the North Pacific. Oral presentation. *Ocean Sciences*. 2024.

Elaina Thomas. Machine learning elucidates the diverse trophic responses of protists to latitudinal and depth-driven changes in their environment. Virtual and in-person poster. *Simons Collaboration on Ocean Processes and Ecology Annual Meeting*. 2023.

Elaina Thomas. Predicted trophic mode of protists across nutrient, temperature, and light conditions of the North Pacific. Oral Presentation. *University of Washington Biological Oceanography Seminar*. 2023.

Elaina Thomas. Predicted trophic mode of protists at the surface and deep chlorophyll maximum of the North Pacific. Oral presentation. *University of Washington Oceanography Graduate Symposium*. 2022.

Elaina Thomas. Predicted trophic mode of protists at the surface and DCM of the subtropical gyre and transition zone. Virtual poster. *Simons Collaboration on Ocean Processes and Ecology Annual Meeting*. 2022.

Elaina Thomas. Mixotrophic protists: sinks or sources of CO_2 across the ocean? Oral presentation. *Graduate Climate Conference*. 2022.

Elaina Thomas, Rika Anderson, Viola Li, Jenni Rogan, Julie A. Huber. Diverse viruses have restricted biogeography in deep-sea hydrothermal vent fluids. Virtual poster. *AGU Fall Meeting*. 2020.

Elaina Thomas, Sean Kearney, Raphaël Laurenceau, Nicolas Raho, and Sallie W. Chisholm. Mechanism behind lanthipeptide gene diversification in marine picocyanobacteria. Virtual poster. *Simons Collaboration on Ocean Processes and Ecology Annual Meeting*. 2020.

Elaina Thomas, Thomas Hackl, Raphaël Laurenceau, Markus J. Ankenbrand, Christina Bliem, Zev Cariani, Paul Berube, Steven J. Biller, Ramunas Stepanauskas, Sallie W. Chisholm. Role of integrative elements in gene transfer and niche adaptation in *Prochlorococcus*. Poster. *Ocean Sciences*. 2020.

Teaching experience

Chemistry of Marine Organic Carbon Teaching Assistant, University of Washington. January–March 2026.
Marine Biogeochemical Cycles Teaching Assistant, University of Washington. March–June 2023.
Statistics Lab Assistant, Carleton College. September 2017–March 2018.
Linear Algebra and Calc. III Grader, Carleton College. September 2017–March 2018.

Mentoring

Lacie Evans, University of Washington Oceanography undergrad. January–June 2026. Through Identity, Belonging and Inquiry in Science (IBIS) Program. "Adaptation of different phytoplankton to varying nitrogen concentrations across ocean basins." Presented poster at IBIS symposium and at University of Washington Undergraduate Research Symposium.
Sage Otulo, University of Washington Oceanography undergraduate. January–June 2025. Through IBIS Program. "Diversity of *Pelagomonas* isolates." Presented poster at IBIS symposium and at University of Washington Undergraduate Research Symposium. Now, undergraduate student worker in ARGO float lab, University of Washington.
Maia Wrice, University of Washington Marine Biology undergraduate. January–June 2024. Through IBIS Program. "Expression of motility-related genes across the diel cycle within an open-ocean protist community." Presented poster at IBIS symposium and at University of Washington Undergraduate Research Symposium. Accepted to present poster at Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS).
Celeste Castañeda-Lopez, University of Washington Marine Biology undergraduate. January–June 2023. Through IBIS Program. "Response of a cosmopolitan mixotrophic plankton (*Chrysochromulina*) to light availability." Presented poster at IBIS symposium and SACNAS poster. Went on to Applied Chemical Science and Technology Masters student, University of Washington.
Anjali Shah, high school student. June–August 2020. I co-supervised, taught her bioinformatics through RNA-Seq analysis of picocyanobacteria lab cultures. Went on to major in Geology-Biology and Applied Mathematics at Brown University.
Veer Gadodia, high school student. June–August 2020. I co-supervised, taught him bioinformatics through RNA-Seq analysis of picocyanobacteria lab cultures. Went on to major in Computer Science at MIT.

Outreach

Presenter, How your data science skills can be used for biological research. Two oral presentations. *University of Washington Youth and Teen Programs Introduction to Math Modeling.* August 6, 2025 and July 23, 2025.

STEM professional mentor, Letters to a Pre-Scientist. 2025, 2021.
Presenter, How your math modeling and data science skills can be used for biological research. Two oral presentations. *University of Washington Youth and Teen Programs Introduction to Math Modeling.* August 22, 2024.
Demo leader, Wedgwood Elementary School Science Night. April 26, 2024.
NSF GRFP peer reviewer, University of Washington**.** 2024, 2023, 2022.
Presenter, Introduction to data science in biology: how I leverage bioinformatics and machine learning to study the trophic mode oceanic protists. Oral Presentation. *University of Washington Data Science in Oceanography Undergraduate Summer Program*. August 8, 2024.
Presenter, University of Washington Data Science in Oceanography Undergraduate Summer Program. August 2023.
Demo leader, University of Washington Aquatic Sciences Open House. May 21, 2023. 582 visitors.
Mentor, Graduate Application Mentorship Program mentor, University of Washington. 2021, 2022.
University of Washington Libraries Storytelling Fellow. July–August 5, 2022.
Science fair mentor, Bryant Elementary School. January–February 2022.
Co-organizer of MIT screening of *Picture a Scientist*. 2020.
Demo leader, MIT Museum Girls Day Exploring Our Watery World. November 2, 2019.

Department service

Banse committee organizer. September 2025–present.
DEI committee member. 2025–present.
Biological Oceanography note taker for faculty meetings. August 2024–present.
Academic and Recreational Graduate Oceanographers. September 2024–present.
Organizer of presentations to prospective graduate students. March 7, 2024.
Helped coordinate Ocean Memory Workshop. 2023.

Fieldwork (49 days at sea)

SCOPE-Gradients 5. Metatranscriptomics and experimental incubations of surface and DCM protist community, with particular focus on trophic mode. R/V Thompson. Chief Scientists E.V. Armbrust and Randelle Bundy. January 21–February 18, 2023.

Biogeochemical Exchanges at Sea Ice Interfaces (BEPSII), Canadian High Arctic Research Station, Cambridge Bay, Nunavut. May 14–23, 2022. Attended lectures on sea ice physics, chemistry, and biology, and practiced field sampling of snow, sea ice, and collected eukaryotic plankton from within and under sea ice.

Return to the Lost City Cruise. Viral metagenomics of a serpentinization-controlled hydrothermal vent system. Lost City Hydrothermal Field. R/V Atlantis. Chief Scientists Susan Lang and William Brazelton. September 8–29, 2018.

Western Cascades, Oregon. Plant-pollinator interactions in alpine meadows. PIs Julia Jones and Rebecca Hutchinson. June–August 2017.

Peer review

Manuscript reviewer for the following journals: Nature Communications, ISME.